

A candidate proof of Problem 5.1 in Covering Complete Geometric Graphs by Monotone Paths

github.com/graph-theory-AI — machine-generated candidate writeup

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Abstract

An odd regular polygon with a central cluster yields point sets requiring at least $3n/4 - O(\sqrt{n})$ crossing-free paths. This candidate proof was generated by GPT-6 Astra in a single model pass for the Graph Theory AI project. It was selected from the campaign because the model marked it `would_publish: true`; it has not been checked by the adversarial referee, a human mathematician, or a novelty search.

1 The problem and the claimed result

This document concerns [Problem 5.1](#) from *Covering Complete Geometric Graphs by Monotone Paths*.

1.1 Problem statement

Problem. Does there exist a constant $c > 1/2$ with the property that, for infinitely many values of n , there are n -element point sets A in the plane such that every covering of the edge set of $K_n[A]$ by crossing-free paths requires at least cn paths?

1.2 Catalog context

Pach, Saghaian, and Schnider proved that every complete convex geometric graph can be decomposed into $n/2 + o(n)$ plane subgraphs. It is open whether a linear lower bound strictly above $n/2$ holds for crossing-free path coverings over general point sets.

1.3 Claimed result

Model verdict: **proved** (confidence: high).

An odd regular polygon with a central cluster yields point sets requiring at least $3n/4 - O(\sqrt{n})$ crossing-free paths.

The model's own caveat: The optimal leading constant is not determined; paths have their standard graph-theoretic meaning.

2 Model writeup

The text below is the model output verbatim, apart from moving its machine-readable verdict block into the summary above and shifting Markdown heading levels for this document. Mechanical TeX defects and equation tags were normalized where needed for compilation.

2.1 An affirmative construction

Write $\pi(A)$ for the minimum number of crossing-free paths covering all edges of $K_n[A]$. Covering paths may overlap, and they need not be Hamiltonian.

We prove the following.

Theorem. 1. For every positive integer q , there is a $6q$ -point set A in general position such that

$$\pi(A) \geq 4q = \frac{2}{3}|A|.$$

2. More strongly, for every sufficiently large n , there is an n -point set A in general position such that

$$\pi(A) \geq \frac{3}{4}n - O(\sqrt{n}).$$

Thus $c = 2/3$ answers the question affirmatively. In fact, every fixed $c < 3/4$ works for all sufficiently large n .

The key is a simultaneous incompatibility between long chords of an odd polygon and edges incident with points near its center. The geometric and counting claims needed below are proved directly.

2.1.1 1. The geometric gadget

Let

$$h = 2m + 1 \geq 5$$

be odd, and let

$$v_j = \left(\cos \frac{2\pi j}{h}, \sin \frac{2\pi j}{h} \right), \quad 0 \leq j < h.$$

Indices are taken modulo h . Put $V = \{v_0, \dots, v_{h-1}\}$, and define the following set of h chords:

$$D = \{v_i v_{i+m} : 0 \leq i < h\}.$$

We need two properties.

2.1.1.1 Property 1: disjoint edges of D cross Fix $v_0 v_m$. Any edge of D disjoint from it has one endpoint in

$$\{v_1, \dots, v_{m-1}\}$$

and the other in

$$\{v_{m+1}, \dots, v_{2m}\}.$$

This follows immediately by checking the indices in $v_i v_{i+m}$. Its endpoints therefore alternate with v_0, v_m around the convex polygon, so the two chords cross. Rotation gives the assertion for every edge of D .

Consequently, every crossing-free path contains at most two edges of D : any three edges of a simple path contain two vertex-disjoint edges. If a crossing-free path contains two edges of D , they must share a vertex.

The two edges of D incident with v_i are precisely

$$v_i v_{i+m} \quad \text{and} \quad v_i v_{i+m+1}.$$

2.1.1.2 Property 2: two incident chords trap a central region Define

$$T_i = \text{conv}\{v_i, v_{i+m}, v_{i+m+1}\}.$$

The origin lies strictly inside every T_i . Indeed, after rotating v_i to $(1, 0)$, the other two vertices have coordinates

$$\left(-\cos \frac{\pi}{h}, \sin \frac{\pi}{h}\right), \quad \left(-\cos \frac{\pi}{h}, -\sin \frac{\pi}{h}\right).$$

Hence

$$U = \bigcap_{i=0}^{h-1} \text{int}(T_i)$$

is an open neighborhood of the origin.

Suppose that a crossing-free path contains

$$v_i a, v_i b, \quad a = v_{i+m}, \quad b = v_{i+m+1}.$$

For any $x \in U$ and any

$$z \in V \setminus \{v_i, a, b\},$$

the segment xz crosses one of these two chords.

To see this, x lies inside T_i , whereas z , being another vertex of a strictly convex polygon, lies outside T_i . The segment xz must leave T_i . It cannot cross ab , since ab is a side of the outer convex polygon and both x and z lie strictly on its inner side. It must therefore cross $v_i a$ or $v_i b$. With the points in general position, this is a proper crossing.

Thus, when both chords are present, edges from central points to outer vertices are possible only at the three vertices v_i, a, b .

2.1.2 2. A particularly simple proof with $c = 2/3$

Take

$$A = V \cup \{o\}, \quad o = (0, 0),$$

and let

$$S = \{ov_j : 0 \leq j < h\}$$

be the central star.

This point set is in general position. Three points on the circle cannot be collinear, and a line through the origin contains two of the polygon vertices only if they are antipodal, which is impossible because h is odd.

We claim that every crossing-free path P satisfies

$$|E(P) \cap (D \cup S)| \leq 3. \quad (1)$$

If P contains at most one edge of D , this follows from

$$\deg_P(o) \leq 2.$$

Otherwise, its two edges of D are $v_i a, v_i b$, as above. Property 2 says that a spoke in P can only be one of

$$ov_i, \quad oa, \quad ob.$$

The edge ov_i is impossible because v_i already has degree two in P . Moreover, oa and ob cannot both occur, since together with $v_i a, v_i b$ they form the cycle

$$o, a, v_i, b, o.$$

A simple path cannot contain a cycle. Hence P contains at most one spoke, proving (1).

Now D and S are disjoint and each has h edges. If k crossing-free paths cover $K_{h+1}[A]$, then, even allowing overlaps,

$$3k \geq |D \cup S| = 2h.$$

Therefore

$$\pi(A) \geq \left\lceil \frac{2h}{3} \right\rceil. \quad (2)$$

Choose $h = 6q - 1$. Then $n = h + 1 = 6q$, and (2) gives

$$\pi(A) \geq \left\lceil \frac{12q - 2}{3} \right\rceil = 4q = \frac{2}{3}n.$$

This already proves the requested assertion.

2.1.3 3. A central cluster improves the coefficient to $3/4 - o(1)$

We now replace the single center by $r \geq 2$ points.

Choose a set $B \subset U$ of r points such that

$$A = V \cup B$$

is in general position. This is possible by choosing the points successively in the open set U , avoiding the finitely many lines determined by previously chosen points.

Let

$$S = \{vx : v \in V, x \in B\},$$

so that

$$|S| = hr.$$

For a crossing-free path P , write

$$d(P) = |E(P) \cap D|, \quad s(P) = |E(P) \cap S|.$$

We have

$$d(P) \leq 2, \quad s(P) \leq 2r. \quad (3)$$

The second inequality follows by summing the path degrees at the r central vertices.

There is a much stronger bound when $d(P) = 2$:

$$d(P) = 2 \implies s(P) \leq 2. \quad (4)$$

Indeed, let the two chords be $v_i a, v_i b$. Property 2 restricts all spokes in P to outer endpoints in $\{v_i, a, b\}$. The vertex v_i already has degree two, while a and b each already have degree one. Thus at most one spoke can use a , and at most one can use b .

Assign weight $2r - 2$ to each edge of D , weight 1 to each edge of S , and weight 0 to every other edge. All other edges remain allowed in the paths.

By (3)–(4), every crossing-free path has weight at most $4r - 2$. Explicitly,

$$(2r - 2)d(P) + s(P) \leq \begin{cases} 2r, & d(P) = 0, \\ (2r - 2) + 2r = 4r - 2, & d(P) = 1, \\ 2(2r - 2) + 2 = 4r - 2, & d(P) = 2. \end{cases}$$

The total edge weight is

$$(2r - 2)|D| + |S| = (2r - 2)h + hr = (3r - 2)h.$$

Summing weights over any covering family therefore proves

$$\pi(A) \geq \left\lceil \frac{(3r - 2)h}{4r - 2} \right\rceil. \quad (5)$$

Finally, for each sufficiently large n , choose an integer

$$r = \sqrt{n} + O(1)$$

such that $h = n - r$ is odd. Then $r \geq 2$ and $h \geq 5$, so the construction applies. Since

$$\frac{3r - 2}{4r - 2} = \frac{3}{4} - \frac{1}{8r - 4},$$

equation (5) yields

$$\begin{aligned} \pi(A) &\geq \left(\frac{3}{4} - \frac{1}{8r - 4} \right) (n - r) \\ &= \frac{3}{4}n - \frac{3}{4}r - \frac{n - r}{8r - 4} \\ &= \frac{3}{4}n - O(\sqrt{n}). \end{aligned}$$

This proves the stronger assertion. The argument covers arbitrary crossing-free paths, including short paths, and allows arbitrary overlap between covering paths. No assumption of a Hamilton-path decomposition is needed.

A Provenance and review status

UNREFEREED MODEL OUTPUT. This document was selected solely because GPT-6 Astra labelled its own result `would_publish: true` and returned `proved` or `disproved`. It has not passed the adversarial LLM referee used for the earlier Sol campaign, has not been checked by a human mathematician, and has not been checked for novelty. Treat every mathematical and bibliographic claim as unverified.

Catalog id	2507.10840__01
Catalog entry	Problem 5.1
Source corpus	arxiv
Campaign leg	selected second attempts (<code>attacks_retry</code>)
Source paper / entry	Covering Complete Geometric Graphs by Monotone Paths
Model verdict	proved (confidence: high)
Selection criterion	<code>would_publish: true</code> and verdict <code>proved</code> or <code>disproved</code>

Model	gpt-6-astra, reasoning effort max, mode=pro, flex service tier
Original artifact	attacks_retry/2507.10840__01/output.md
Independent review	None
arXiv	arXiv:2507.10840
Previous attempt	attacks/2507.10840__01: gpt-5.6-sol verdict partial
Model's caveats	The optimal leading constant is not determined; paths have their standard graph-theoretic meaning.

The generated Markdown and LaTeX sources for this PDF are stored under to_review_astra/src/2507.10840__01/.